Tianwei Jiang

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Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Electrical and Computer Engineering – Digital Signal Processing — *Aug. 2024 – Present*

- **GPA:** 4.0/4.0
- **Research Interests:** Audio Deep Learning, Multimodal Learning, Audio Perception, Sound Event Detection, Sound Source Localization, Audio-Visual Learning, Spatial Audio, and Audio Foundation Models

University of Rochester — Rochester, NY
B.S. in Audio and Music Engineering; B.A. in Mathematics — *Aug. 2018 – May 2022*

- **Overall GPA:** 3.84/4.0; **Audio and Music Engineering GPA:** 3.97/4.0
- Professor's Choice Award, Undergraduate Research Fair, Dean's List every semester

First-Author Publications

- **Tianwei Jiang**, et al. *Human-in-the-Loop Active Learning for Rare Poultry Vocalization Discovery in Long-Term Farm Recordings.* Published in IEEE MLSP 2026.
- **Tianwei Jiang**, et al. *Modality Compensation via Explicit Audio-Visual Correspondence for Multi-Source Sound Localization.* Submitted to ICASSP 2027.
- **Tianwei Jiang**, et al. *Automatic Discovery of Mixed Call Types and Soft-Label Learning for Pekin Duck Vocalization Classification* Submitted to ICASSP 2027.
- **Tianwei Jiang**, et al. *Generalizable HRTF Elevation Modeling Using Relative Binaural Magnitude and Group Delay* Submitted to ICASSP 2027. *In collaboration with Dolby Labs.*
- **Tianwei Jiang**, et al. *Elevated Sources and HRTFs: A Boundary Element Study of Ear Shape and Spectral Cues* Under review at The Journal of the Acoustical Society of America (JASA). *In collaboration with Dolby Labs.*

Research Experience

Georgia Institute of Technology — Atlanta, GA
Graduate Research Assistant, Advisor: Prof. David Anderson — *Aug. 2024 – Present*

- **Multimodal Machine Learning for Audio Perception.** Investigating explicit compensation mechanisms across audio, vision to improve interpretability and robustness. Developed a multi-source sound localization framework that uses candidate-conditioned audio-visual reasoning to associate acoustic sources with visual objects and refine localization.
- **Multimodal Perception for Parkinson's Disease.** In collaboration with the joint Georgia Tech–Emory Biomedical Engineering program, investigating speech, visual, and body movement data to identify behavioral and motor cues that may support early diagnosis of Parkinson's disease. Developing a speaker diarization pipeline to quantify the frequency of patients' conversational participation.
- **Sound Event Detection for Bioacoustics.** Developed an end-to-end pipeline for Pekin duck vocalization classification by introducing soft-target supervision to identify and assign mixture vocalizations, addressing challenges of short-duration calls, highly similar acoustic patterns, ambiguous labels, and limited training data.
- Designed a human-in-the-loop active learning framework for rare chicken vocalization discovery in long-term farm recordings by iteratively mining hard negative samples and positive target events, substantially improving dataset quality and classifier performance with minimal annotation effort.

Industry Experience

Dolby Laboratories — San Francisco, CA/ Atlanta, GA
Audio Software Engineer Intern — *May 2025 – May 2026*

- Owned end-to-end sound source localization R&D, improving median-plane elevation estimation and developing height localization algorithms toward IP and publication.
- Investigated the distinct contributions of the helix and tragus to elevation cues through acoustic simulations and controlled ear-structure ablations, analyzing frequency-dependent changes in spectral peaks, notches, and interaural differences.

- Designed binaural and monaural features informed by ear acoustics, including interaural time and level differences, group delay, and pinna-induced spectral cues; trained compact CNN and MLP models for robust elevation estimation.
- Analyzed multiple HRTF datasets using PCA, UMAP, and layer-wise diagnostics to characterize elevation-dependent representations and their relationship to pinna filtering mechanisms.
- Built an end-to-end evaluation pipeline and integrated the validated algorithm into internal binaural and 7.1.4 spatial rendering pipelines; IP filing in progress.

Agora Inc. Shanghai, China
Audio Lab Engineer *Jan. 2023 – Aug. 2024*

- Developed automated testing systems to ensure SDK quality in collaborative software development environments.
- Designed Raspberry Pi-based hardware routing automation and Appium-based iOS and Android testing pipelines.
- Implemented POLQA, ViSQOL, and PESQ evaluation systems for audio quality verification.

Graduate Projects

Localized Mode Degradation Modeling for Contact Acoustics

Research Project with Robotics and Areospace Engineering *Jan.2026 - Present*

- Developed a localized mode degradation algorithm to model how individual resonant modes change under real-world contact and material conditions, rather than applying a single global transformation to the full spectrum.
- Designed a degradation encoder to estimate mode-specific parameters, including amplitude attenuation, frequency shift, bandwidth variation, decay, temporal displacement, onset smearing, noise floor, and spectral tilt.
- Constructed a simulation-to-real adaptation pipeline that transforms clean synthetic acoustic responses into realistic degraded signals while preserving class-dependent modal structure.
- Analyzed within-class and between-class spectral variation to identify stable contact-response patterns and separate material-specific characteristics from variations in impact strength.

Biomarker Detection Using Vision Transformers

Course Project *Oct. 2024 – Nov. 2024*

- Implemented ViT and EVA-02 models with ensemble learning for biomarker detection on the OLIVES dataset.
- Achieved an F1 score of 0.87, outperforming all baseline methods.

Undergraduate Research

"Singing Glasses": Digital Adaptation of the Wine Glass Effect

University of Rochester
Undergraduate Researcher, Advisor: Prof. Michael Heilemann *May 2021 – May 2022*

- Collected impulse responses using piezo pickups and designed digital envelopes to simulate the Glass Armonica.
- Developed a reverb plug-in and MIDI-controlled robotic performance system using Rumba boards.
- Received the Undergraduate Discover Grant and Professor's Choice Award.

Skills

Programming: Python, MATLAB, C/C++, PyTest

Machine Learning: PyTorch, Vision Transformers, CNNs, Transfer Learning, Audio Machine Learning, Multimodal Learning

Audio and DSP: Sound Source Localization, Sound Event Detection, Spatial Audio, HRTFs, Max/MSP, JUCE

Audio Engineering: Pro Tools, Logic Pro X, Cubase, Reaper, Recording, Mixing, Microphone Techniques, and DSP Plug-ins